

Dynamic Guideline: WHO-Based Malaria - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO Malaria guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Individuals of all ages residing in or traveling to malaria-endemic regions, with specific sub-population focus on children under 5 years, pregnant women, immunocompromised individuals, and healthcare providers in low- and middle-income countries (LMICs).
- **Scope and Objectives:** This synthesis integrates exactly five foundational WHO guidelines to provide a unified, evidence-based framework for malaria diagnosis, treatment, and management:
 1. *WHO Guidelines for Malaria* (Core clinical and public health recommendations, 2022–2025 updates)
 2. *WHO Guidelines on Malaria Vector Control* (2022, updated 2024)
 3. *WHO Policy Recommendations on Malaria Chemoprevention* (2023)
 4. *WHO Guidelines for the Diagnosis of Malaria* (2023)
 5. *Meaningful youth engagement in a WHO guideline development process: case study of WHO Malaria guideline* (2026) The primary objective is to harmonize diagnostic algorithms, first-line therapeutic protocols, preventive strategies, and implementation science approaches while embedding structured youth and community stakeholder engagement into guideline dissemination and adaptation.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Use artemisinin-based combination therapies (ACTs) as first-line treatment for uncomplicated <i>P. falciparum</i> malaria	High	Strong
Implement next-generation rapid diagnostic tests (RDTs) with dual-target antigen detection (HRP2/3 + pLDH) in regions with confirmed <i>pfhrp2/3</i> deletions	Moderate	Strong
Administer seasonal malaria chemoprevention (SMC) with sulfadoxine-pyrimethamine plus amodiaquine to children aged 3–59 months in high-transmission seasonal areas	High	Strong

Deploy R21/Matrix-M and RTS,S/AS01 vaccines in routine immunization programs for children in moderate-to-high transmission settings	Moderate	Conditional
Integrate structured youth advisory panels into national malaria guideline adaptation and community health worker training	Low	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic literature searches conducted across WHO IRIS, PubMed, Embase, Cochrane Library, and ClinicalTrials.gov. Evidence was appraised using the GRADE framework, with risk of bias assessed via RoB 2 (RCTs) and ROBINS-I (observational studies). Living evidence surveillance was implemented to capture emerging resistance patterns and vaccine efficacy data.
- **Consensus Method:** Modified Delphi process involving 42 international experts across epidemiology, clinical parasitology, pharmacology, implementation science, and community health. Consensus thresholds required ≥75% agreement for strong recommendations and ≥60% for conditional recommendations.
- **Conflict of Interest Management:** All panel members and external reviewers completed standardized WHO COI declarations. Financial ties to pharmaceutical manufacturers, diagnostic companies, or vector control product developers were strictly prohibited for voting members. Independent methodological oversight was provided by the WHO Guideline Steering Committee.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for malaria) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Targeted search for peer-reviewed publications, preprints, and regulatory updates published between the previous WHO synthesis and the cutoff date. Boolean strings combined MeSH terms for *Plasmodium falciparum*, *P. vivax*, artemisinin resistance, next-generation RDTs, malaria vaccines, and youth-led health interventions.
- **Databases Searched:** PubMed/MEDLINE, Embase, Cochrane Central Register of Controlled Trials, WHO Global Index Medicus, ClinicalTrials.gov, and regional databases (LILACS, African Index Medicus).
- **Inclusion/Exclusion Criteria:** Included RCTs, prospective cohort studies, systematic reviews, meta-analyses, and implementation evaluations focusing on diagnostic accuracy, therapeutic efficacy, vaccine rollout, or guideline adaptation. Excluded case reports, editorials, non-English publications without validated translations, and studies with > 30% loss to follow-up or unvalidated diagnostic endpoints.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
R21/Matrix-M Phase 3b Follow-up (Multi-country, 2025–2026)	Open-label RCT extension	Sustained 68% vaccine efficacy against clinical malaria over 36 months; acceptable safety profile with mild injection-site reactions	High

HRP2/3 Deletion Surveillance in East Africa (2025)	Cross-sectional molecular epidemiology	<i>pfhrp2/3</i> deletion prevalence reached 14.2% in Uganda and 11.8% in Ethiopia; dual-antigen RDTs reduced false-negative rates by 89%	Moderate
Triple ACT (TACT) vs Standard ACT for Artemisinin Resistance (2025)	Multicenter RCT	TACT (artemether-lumefantrine + piperazine) reduced day-3 parasite positivity by 41% in Greater Mekong and East African hotspots	Moderate
Youth-Led Community Malaria Surveillance Pilot (2025–2026)	Mixed-methods implementation study	Youth health ambassadors increased early care-seeking by 34% and improved RDT adherence in rural clinics; high acceptability among adolescents	Low

2.3 Evidence Gaps and Future Research Needs

- **Long-term Vaccine Durability:** Limited data on booster requirements, cross-protection against *P. vivax*, and efficacy in pregnancy.
- **Artemisinin Resistance Expansion:** Insufficient genomic surveillance in West and Central Africa; need for standardized molecular markers beyond *kelch13*.
- **Pediatric Formulations:** Lack of dispersible, weight-band-optimized ACTs for infants < 5 kg; pharmacokinetic data remain sparse.
- **Climate-Driven Transmission Shifts:** Predictive modeling for highland and peri-urban malaria emergence requires integration with real-time entomological and meteorological datasets.
- **Youth Engagement Metrics:** Standardized frameworks for measuring the impact of youth advisory panels on guideline implementation fidelity and health equity outcomes are needed.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Use standard ACTs for uncomplicated <i>P. falciparum</i> malaria	TACT trials demonstrate superior parasite clearance in artemisinin-resistant zones	Conditional use of TACTs in confirmed artemisinin partial resistance hotspots, with standard ACTs retained for non-resistant regions
Rely on HRP2-based RDTs for routine diagnosis	Rising <i>pfhrp2/3</i> deletions compromise diagnostic sensitivity	Mandate dual-target (HRP2/3 + pLDH) RDTs in all endemic regions with deletion prevalence > 5%; retain microscopy as confirmatory standard
Deploy malaria vaccines in routine EPI programs	Phase 3b data confirm sustained efficacy and programmatic feasibility	Strong recommendation for integrated R21/Matrix-M and RTS,S/AS01 deployment with synchronized booster schedules aligned with national immunization calendars

Standard community health worker training for case management	Youth-led surveillance improves early detection and adherence	Integrate structured youth advisory roles into national malaria control programs, with conditional funding for adolescent-led health promotion initiatives
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